

positioning is not just about selecting a high-performance GNSS receiver. It depends on choosing the right correction method for the application.

Standard GNSS systems, such as the Global Positioning System, Galileo, and GLONASS, provide accuracies of two to five metres. This level of precision is suitable for tracking but not for precise control.

Differential global positioning system (DGPS) improves accuracy to the sub-metre level. It works well for mapping and maritime use. However, it cannot reliably achieve centimetre-level precision.

For precision agriculture, surveying, drones, and robotics, real-time kinematic (RTK) is widely used. RTK relies on a base station and a rover. The base station sends carrier-phase corrections in real time via the internet, radio, or cellular links.

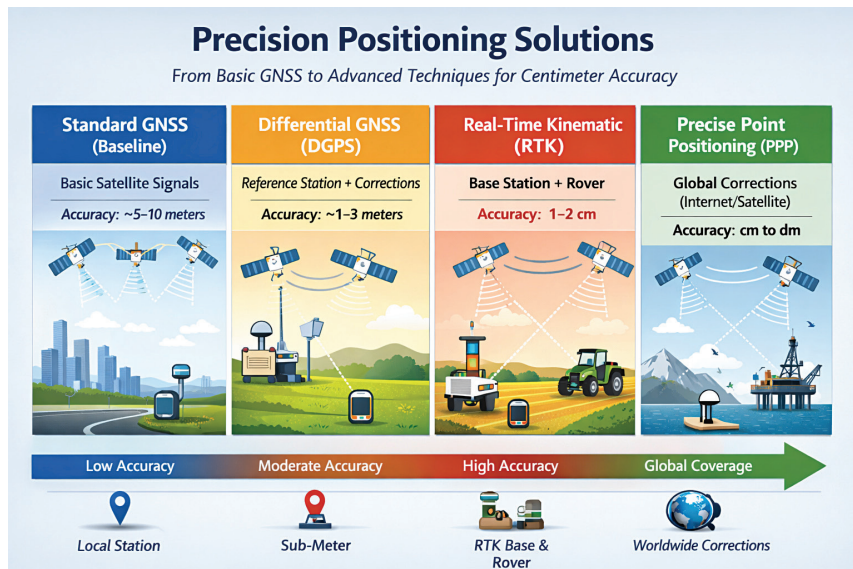
This enables fast, multi-band positioning with hardware such as the ZED-F9P from u-blox. It is commonly integrated into systems from John Deere and DJI.

The main drawback is the requirement for continuous correction connectivity and supporting local infrastructure.

Precise point positioning (PPP) removes the need for a local base station. It uses precise satellite orbit and clock corrections delivered via satellite or the internet from global networks.

After a convergence period ranging from several minutes to about 30 minutes, PPP achieves centimetre-to-decimeter accuracy worldwide. This makes it suitable for offshore operations, mining, and global fleets. Examples include Trimble RTX and Topcon correction services.

However, PPP requires longer initialisation times, reliable connectivity, and higher processing power. Hybrid PPP-RTK combines



global corrections with fast ambiguity resolution similar to RTK. It offers quicker convergence than standalone PPP while maintaining wide-area scalability.

Achieving centimetre-level accuracy requires balancing correction strategy, communication

reliability, convergence time, scalability, and system complexity to match mission requirements.

Solution B. Selecting the right GNSS architecture. Choosing the right GNSS architecture is critical because it directly affects accuracy, cost, and overall system performance.

The first step is to determine whether the application requires real-time centimetre-level accuracy or whether delayed corrections are acceptable. RTK provides immediate centimetre-level positioning and is widely used in precision agriculture, robotics, and autonomous vehicles. However, it requires reliable connectivity, low-latency correction data, and clear satellite visibility.

PPP can achieve similar accuracy without a local base station. It uses satellite-delivered corrections but requires several minutes for convergence. PPP works well for long-term asset tracking, mapping, and surveying.

Environmental conditions strongly influence GNSS performance. Urban

